# Air Drawing Physics — Complete Guide

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🎨 Draw in the air with your finger, then turn your drawings into real physics objects. This guide shows every feature and the exact steps to reach a goal.

## 🚀 Quick start (5 minutes)

1. **Start DroidCam** on your phone, then run `python air_drawing.py`. A window opens with your camera view as the background. 📷
2. **Show your right hand** to the camera and point with **only your index finger** ☝️. A coloured trail follows your fingertip: you are drawing.
3. **Change colour:** hold your fingertip on a colour square (top right) for half a second. 🎨
4. **Make it real:** open your whole right palm 🖐 (or press `Space`). Your drawing becomes a physics object that falls and bounces.
5. **Move it:** pinch **thumb + middle finger** 🤏 on an object to grab it, move your hand, open to throw it.
6. **Undo a mistake:** press `Z`. **Clear everything:** press `C`.
7. **Forgot something?** Press `Ctrl` + `/` any time for the quick guide on screen. 📖

💡 **The golden rule of every button:** hold your fingertip on it for **0.5 seconds**. A bar fills along the bottom of the button, then it fires. Moving away early cancels it.

## 🖐 Hand gestures — the cheat sheet

| Hand | Gesture | What it does |
|---|---|---|
| Right | ☝ Index finger only (others folded) | **Draw** |
| Right | 🤏 Thumb + index pinch, other fingers folded | **Pen size** — pinch closer = thinner, wider = thicker |
| Right | ✊ Fist | **Erase** (a circle follows your fist) |
| Right | 🖐 Open palm, hold briefly | **Launch** drawings into physics |
| Right | 🖐 Open palm, hold 1 second | **Radial menu** opens around your hand |

| Hand | Gesture | What it does |
|---|---|---|
| Either | 🤏 Thumb + middle finger pinch | **Grab** an object: move, throw, catch |
| Left | ✊ Fist | **Push** nearby objects away |
| Left | 🖐 Open palm | **Blast** — an explosion at your hand |
| Both | ☝☝ Both index fingers (turn on with `W`) | **Two-hand drawing** |

### 🧠 Tips for steady tracking

- Keep your hand fully inside the camera view and well lit. 💡
- Make gestures clearly: a half-folded finger can read as the wrong gesture.
- Gestures need 2–3 frames to register, so hold each one for a moment.

## 🖥️ The screen — where everything lives

The screen has four zones. Every button is pressed by **holding a fingertip on it for 0.5 s** (or clicking with the mouse).

| Zone | Where | What is there |
|---|---|---|
| 🎨 Colours | Top right | 8 colour squares (keys `1` – `8`) |
| 📑 Tabs | One row under the top edge | FILE, PLAY, ART, UI, AI, TIME, LAYERS, EDIT, FX, BRUSH |
| 🧰 Toolbar | Bottom | 16 buttons, all the same size |
| 📊 Info card | Left, under the tabs | Gravity, trails, pen size (hides while a tab is open) |

**The tabs** (only one is open at a time):

- 📁 **FILE** — export, save/load projects, record video
- 🎮 **PLAY** — games
- 🪞 **ART** — two-hand, symmetry, generative art
- 🎛️ **UI** — themes, HUD, modes, guide
- 🧠 **AI** — recognise, clean up, assistant, calculator

- 🕒 **TIME** — history, jump back to any step
- 🗂️ **LAYERS** — show, hide, lock, reorder
- ✏️ **EDIT** — properties of the selected drawing
- ✨ **FX** — particle effects
- 🖌️ **BRUSH** — 12 brushes

**The bottom toolbar:** LAUNCH · CLEAR · UNDO · REDO · GRAVITY · TRAILS · BRUSH · <WIND · WIND> · PHOTO · BLAST · TO 3D · MODE 2D/3D · FX · MODE (draw/select/…) · THEME.

🚫 **No accidental ink:** while your finger is over a tab, panel, colour square or toolbar button, drawing is paused, so reaching for a button never leaves marks. The bottom buttons also react when your finger is at the very bottom edge of the camera view.

## ✏️ Drawing — brushes, size, two hands

**Choose a brush** 🖌️: open the **BRUSH** tab, hold your finger on a row. Or press `Tab` to cycle, or `F1` – `F12` to jump straight to one.

| Key | Brush | Look |
|---|---|---|
| F1 | Pencil | thin grey sketch line |
| F2 | Marker | solid, slightly wide |
| F3 | Crayon | rough, waxy |
| F4 | Pen | clean ink |
| F5 | Neon | glowing tube ✨ |
| F6 | Fire | flames 🔥 |
| F7 | Ice | frosty ❄️ |
| F8 | Lightning | electric ⚡ |
| F9 | Particle | spray of dots |
| F10 | Galaxy | stars and glow 🌌 |
| F11 | Water | flowing blue 💧 |
| F12 | Smoke | soft cloud |

**Pen size:** pinch thumb + index (right hand, other fingers folded) or press `+` / `-`.

**Colour:** hold on a colour square, or press `1` (red) to `8` (black).

## 🙌 Two-hand drawing

Draw with **both index fingers at the same time**.

1. Press `W` (or ART tab → TWO-HAND). A message confirms it is on.
2. Point with the **index finger of each hand**. Each hand draws its own stroke.
3. Your **right hand uses your chosen colour**; your **left hand uses the opposite (complementary) colour** so you can tell them apart.
4. Lift a finger to finish that hand's stroke. Each stroke is its own undo step ( `Z` removes the newest).
5. Press `W` again to turn it off. With it off, the left hand does not draw.

💡 Two-hand drawing combines with symmetry ( `M` ) and generative art ( `J` ). It is off in 3D mode.

# 🌍 Physics and 3D

## 🚀 Launch, grab, throw

1. Draw one or more strokes. They are just ink at this point.
2. **Launch:** open your right palm 🖐 or press `Space` or press LAUNCH. Every stroke becomes a rigid body with gravity. 💥
3. **Grab:** pinch thumb + middle finger 🤏 on an object. It follows your hand. Open your fingers to **throw** it with the speed of your hand.
4. While holding: `[` / `]` make it smaller / bigger, `P` / `O` pull / push it.
5. **Left fist** ✊ pushes things away. **Left open palm** 🖐 explodes the area (also `E`).

**World controls:** `G` cycles gravity (down / up / zero) · `T` shows motion trails · the <WIND and WIND> buttons blow objects sideways.

## 🧊 3D mode

- Press `X` (or MODE 2D/3D). Drawings become solids with depth.
- **Move your hand toward the camera** = the stroke gets closer.
- **Left fist drag** = orbit the view. Arrow keys rotate; `PageUp` / `PageDown` change depth; `0` resets depth and view.
- `V` turns the last flat drawing into a 3D solid (a circle becomes a sphere, a square a cube, and so on). `N` makes this automatic.
- Grab a 3D object with the same thumb + middle pinch. Hand it from one hand to the other and it follows.

## ✨ Effects

Open the **FX** tab and hold on a row to attach an effect to the drawing you hold or just drew: Fire, Smoke, Sparks, Snow, Stars, Electricity, Plasma, Dust, Magic, Confetti or Bubbles. On a drawn **circle**, an effect becomes an **energy ring**.

# 🧠 AI, calculator, symmetry and generative art

## 🧠 AI tab ( `A` )

The AI works with **geometry rules and templates on your computer** — no internet, no machine learning. It looks at a finished drawing about one second after you stop.

| Button | What it does |
|---|---|
| RECOGNIZE | Names the drawing (circle, square, triangle, star, heart, house, tree, car, person, arrow, cloud, sun, spaceship, galaxy…) |
| CLEAN UP | Replaces your wobbly shape with a perfect one |
| ENHANCE | Smooths and improves the sketch |
| AUTO CLEAN | Cleans every shape you draw automatically |
| ASK `/` | Type a command in the ask box, press Enter |

💬 **Assistant examples** (press `/`): "draw a house", "make it red", "make it bigger", "move it left", "delete this", "what is this?", "use the neon brush". ⌨️ Press `Esc` to close the box. Every AI edit can be undone with `Z`.

## 🧮 Air calculator (`K`)

1. Press `K` (CALC). Write a sum with your finger in the air, for example **25+30**.
2. Pause for about 1.5 seconds. The answer appears next to your writing.
3. Press `K` again for **MATH** mode, which handles longer expressions.

## 🪞 Symmetry (`M`, ART tab)

Press `M` to cycle: **off → horizontal → vertical → radial → kaleidoscope**. You see the mirrored lines live while you draw. When you lift your finger, the copies become real strokes.

- **FOLDS** sets how many copies radial / kaleidoscope make (2–16).
- **CENTER** is the screen middle, or the middle of your last stroke (button LAST).
- The whole group is **one undo step**.

## 🌸 Generative art (`J`, ART tab)

Turn **J** on, then draw one simple stroke. It grows a pattern:

- a **line** → a fractal tree 🌳
- a **circle** → a mandala 🌼
- a **triangle** → a Sierpinski triangle 🔺

- anything else → a rosette

Sliders: STRENGTH (detail), PETALS/FOLDS, ROTATION, PARTICLES. Already drew something? Press **GENERATE** to use your last stroke. One undo removes the whole pattern.

## 🎮 Games, save, export and record

### 🎮 Games (PLAY tab or `Ctrl` + `G`)

When a game starts, your current drawing is **saved and the board is cleared**. When you press STOP, your drawing **comes back**.

| Game | How to play | Score |
|---|---|---|
| 🎯 TARGETS | Draw a shape, launch it, make it touch the glowing targets. 45 seconds. | 100 per hit, bonus for combos |
| ✏️ DRAW IT | Draw the named shape in **one stroke**. 5 rounds. | Accuracy 50%, speed 20%, smoothness 15%, completion 15% — reasons shown on screen |
| 🌉 BRIDGE | Draw a bridge across the gap, launch it (`Space`). It locks in place and the ball rolls to the flag. | 1000, minus 100 per retry and bridge length |

NEXT / RETRY moves on, STOP leaves. Best scores are kept in `projects/game_scores.json`.

### 💾 Save and open projects

- `Ctrl` + `S` or FILE → SAVE writes a `.airdraw` file to the `projects` folder.
- FILE → LOAD LATEST, or tap one of the three newest files listed, opens it. A damaged file is refused with a message and nothing changes.
- **AUTOSAVE** (off by default) saves every 30 s and keeps the newest 10 autosaves. It never overwrites your named saves.

### 🖼️ Export (FILE tab, folder `exports`)

| Button | Result |
|---|---|
| PNG | 1280×720 image of your art without menus |
| JPG | the same as a JPG |
| SVG | vector lines (strokes only; 3D solids and particles are not included) |
| JSON | list of strokes with points, colours, widths |
| BG | background: DARK, your THEME, or CLEAR (transparent, PNG only) |

📸 The toolbar **PHOTO** button (or `S`) saves a screenshot of everything you see.

### 🔴 Record

- `R` or FILE → MP4 starts recording the screen (up to 120 s). Press again to stop. If your OpenCV has no MP4 encoder, an AVI is saved instead.
- **GIF** records a short looping GIF (8 frames per second).
- Files appear in `exports`; a red REC marker shows at the top while recording.

## 🗂️ Layers, edit, history, themes and modes

### 🗂️ LAYERS (`L`)

Five layers: Background, Drawing, Physics Objects, Effects and UI. Each has two rows:

- Row 1: **select** the layer · **SHOW / HIDDEN** · **LOCK** (stops edits on it).
- Row 2: **opacity** − / + · **UP / DN** (reorder) · **CLR** (clear that layer).
- `Ctrl` + `R` renames the selected layer.

### ✏️ EDIT (`I`)

Pinch an object (or pick one in SELECT mode), then open EDIT to change its **colour, brush, size, opacity, glow, effect, mass, bounce, friction, gravity**, freeze it, hide it, **DUPLICATE** or **DELETE** it.

## 🕒 TIME ( `H` )

Every action is logged. Hold a finger on any entry to **jump back to that moment**. UNDO / REDO sit at the top ( `Z` , `Y` , `Ctrl` + `Y` ).

## 🌌 Themes (UI tab or `Ctrl` + `T` )

| Theme | Look |
|---|---|
| CAMERA | your camera picture (default) |
| CYBER | neon grid city |
| SPACE | stars and nebula |
| LAB | clean technical grid |
| HOLO | hologram |

## 📟 HUD ( `` ` `` )

Cycles **classic** (info card + bottom bar) → **system** (a futuristic dashboard) → **off**.

## 🎛️ Modes ( `Ctrl` + `1` … `9` , or the MODE button)

| Key | Mode | Meaning |
|---|---|---|
| Ctrl+1 | DRAW | point to draw |
| Ctrl+2 | SELECT | hover an object to select it (no ink) |
| Ctrl+3 | GRAB | move objects (no ink) |
| Ctrl+4 | EDIT | opens the EDIT panel |
| Ctrl+5 | PHYSICS | each finished stroke launches at once |
| Ctrl+6 | 3D | depth drawing |
| Ctrl+7 | AI | opens the AI panel |
| Ctrl+8 | GAME | opens the PLAY panel |
| Ctrl+9 | SETTINGS | opens the UI panel |

# ⌨️ Every keyboard shortcut

| Key | Action |
|---|---|
| `Space` | Launch drawings into physics |
| `C` | Clear everything |
| `Z` / `Ctrl` + `Z` | Undo |
| `Y` / `Ctrl` + `Y` | Redo |
| `G` | Gravity: down / up / zero |
| `T` | Motion trails |
| `E` | Explosion |
| `S` | Screenshot |
| `B` | Brush tab |
| `Tab` / `Shift` + `Tab` | Next / previous brush |
| `F1` – `F12` | Pick a brush |
| `F` | FX tab |
| `I` | EDIT tab |
| `L` | LAYERS tab |
| `H` | TIME (history) tab |
| `A` | AI tab |
| `/` | Ask the assistant |
| `K` | Air calculator (press again for math) |
| `U` | UI tab |
| `` ` `` | HUD: classic / system / off |
| `1` – `8` | Colours |
| `+` / `-` | Pen size |
| `[` / `]` | Shrink / grow the grabbed object |

| Key | Action |
|---|---|
| `P` / `O` | Pull / push the grabbed object |
| `X` | 2D / 3D mode |
| `V` | Turn last drawing into 3D |
| `N` | Auto-3D on / off |
| Arrow keys, `PageUp`, `PageDown`, `0` | 3D rotate, depth, reset |
| `D` | Gesture debug overlay |
| `W` | Two-hand drawing |
| `M` | Symmetry modes |
| `J` | Generative art |
| `R` | Record MP4 |
| `Ctrl` + `S` | Save project |
| `Ctrl` + `G` | Game mode on / off |
| `Ctrl` + `T` | Next theme |
| `Ctrl` + `1` – `9` | Switch mode |
| `Ctrl` + `R` | Rename layer |
| `Ctrl` + `/` | Quick guide on screen |
| `Q` / `Esc` | Quit (Esc first closes the guide or the ask box) |

## 🎯 I want to… — task finder

| I want to… | Do this |
|---|---|
| Draw a line | ☝ Point with only your right index finger |
| Draw a perfect circle or square | Draw it roughly, then AI tab → CLEAN UP (or turn on AUTO CLEAN) |
| Draw with both hands | Press `W`, point with both index fingers |

| I want to… | Do this |
|---|---|
| Draw a mirror image | Press `M` until HORIZ / VERT / RADIAL / KALEID, then draw |
| Make a mandala or tree from a doodle | Press `J`, draw a circle (mandala) or a line (tree) |
| Change colour | Hold finger on a colour square, or press `1`–`8` |
| Change brush | BRUSH tab, or `Tab`, or `F1`–`F12` |
| Make a glowing neon line | `F5` (Neon) |
| Make drawings fall and bounce | `Space` or open right palm |
| Move or throw something | 🤏 Thumb + middle pinch on it, move, release |
| Blow things away | Left fist (push) or left open palm (blast) |
| Attach fire, snow or sparks | FX tab → hold on an effect |
| Make it 3D | `X`, or draw then press `V` |
| Ask the computer to draw something | `/` then type "draw a house" |
| Do a quick calculation | `K`, write `25+30` in the air, wait |
| Play a game | PLAY tab → TARGETS, DRAW IT or BRIDGE |
| Change the look of the screen | `Ctrl` + `T`, or UI tab → theme |
| Save my work to continue later | `Ctrl` + `S` (open it from FILE → LOAD) |
| Get a picture of my art | FILE → PNG / JPG / SVG |
| Record a video | Press `R` (stop with `R`), or FILE → GIF |
| Fix a mistake | `Z` for undo, TIME tab to jump back further |
| Hide or lock part of my scene | LAYERS tab |
| Change one object's bounciness or colour | Pinch it, then EDIT tab |
| Start over | `C` |
| See all controls on screen | `Ctrl` + `/` |

# 🛠️ Troubleshooting

| Problem | Fix |
|---|---|
| No camera picture | Check DroidCam is running and the phone and PC share the same Wi-Fi; the app uses the DroidCam address already set in the file |
| Hand not detected | Better light, keep the whole hand in view, avoid a busy background |
| Draws when I do not want it | Fold the other fingers clearly; a loose hand reads as another gesture |
| Button does not fire | Hold still on it for the full 0.5 s until the bar fills; a short tracking drop-out is forgiven for 0.18 s |
| Cannot reach the bottom buttons | Move your finger to the bottom of the camera view — the hit area reaches the very bottom edge |
| Tab will not open | Only one tab is open at a time; the new tab closes the old one |
| Left hand does nothing | Left hand draws only when two-hand mode is on ( `W` ) |
| Game cleared my drawing | It is restored when you press STOP |
| MP4 saved as AVI | Your OpenCV has no MP4 encoder; the AVI plays normally |
| Choppy video | Turn the HUD off ( `` ` `` ), lower the theme effects (UI tab → ANIM), close other programs |

📌 **Honest limits:** shape recognition is rule-based, not machine learning; symmetry is 2D only; SVG holds strokes only; the autosave default is off.